Lie Theory: Chapter 3 Homework

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1. Let L be a non-abelian 3-dimensional Lie algebra with basis $\{f, g, z\}$ and $[f, g] := z$. Assume $z \in Z(L)$ and L' is one-dimensional. Prove that $L' = Z(L)$.

2. Check that the upper triangular matrices $\{e_{12}, e_{23}, e_{13}\}$ satisfy the following properties.

(a) $[e_{12}, e_{23}] = e_{13}$.

(b) e_{13} is in the center of the Lie algebra of upper triangular matrices.

This shows that the set $\{e_{12}, e_{23}, e_{13}\}$ satisfies the construction described in Section 3.2.1 with $z = e_{13}$ and f and g equal to e_{12} and e_{23}.

3. Let V be a vector space and let φ be an endomorphism of V. Let L have underlying vector space $V \oplus \mathrm{Span}\{x\}$. Show that if we define the Lie bracket on L by $[y, z] = 0$ and $[x, y] = \varphi(y)$ for $y, z \in V$, then L is a Lie algebra and $\dim(L') = \mathrm{rank}(\varphi)$.

4. Suppose L is a vector space with basis $\{x, y\}$ and that a bilinear operation on L is defined such that $[u, u] = 0$ for all $u \in L$. Show that the Jacobi identity holds and hence L is a Lie algebra.

5. Suppose that I is an ideal of a Lie algebra L and that there is a subalgebra S of L such that $L = S \oplus I$. Show that the map $\theta : S \to \mathfrak{gl}(I)$ defined by $\theta(s)(x) = [s, x]$ is a Lie algebra homomorphism from S into $\mathrm{Der}(I)$.

We say that L is a *semidirect product* of I by S.

6. Show conversely that given Lie algebras S and I and a Lie algebra homomorphism $\theta : S \to \mathrm{Der}(I)$, the vector space $S \oplus I$ may be made into a Lie algebra by defining

$$[(s_1, x_1), (s_2, x_2)] = ([s_1, s_2], [x_1, x_2] + \theta(s_1)x_2 - \theta(s_2)x_1)$$

for $s_1, s_2 \in S$, and $x_1, x_2 \in I$, and that this Lie algebra is a semidirect product of I by S. (The direct sum construction introduced in Exercise 2.6 is the special case where $\theta(s) = 0$ for all $s \in S$.)